A Design of a Grey-Predicted Li-Ion Battery Charge System

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***Abstract*—In this paper, we consider an Li-ion battery as a grey system. The grey prediction technique is then used to develop a grey-predicted Li-ion battery charge system (GP-LBCS). The proposed GP-LBCS is designed to replace the general constant-voltage (CV) mode using the GP mode to improve the Li-ion battery charge behavior. A GP algorithm is built in GP-LBCS to make the charge trajectories faster and safer. A 3-D Y-mesh diagram for describing the charge trajectories of the proposed GP-LBCS is simulated. A GP-LBCS prototype is designed and implemented to assess the charge performance. Experiments show that the charge speed and the charge efficiency of the proposed GP-LBCS, compared with the general constant-current–CV charge system, are increased above 23% and 1.6%, respectively. The charge speed and efficiency in the proposed GP mode are improved above 34% and 7%, respectively, compared with that in the general CV mode.**

***Index Terms*—Battery charge system, grey-predicted (GP), Li-ion battery.**

I. Introduction

The development of mobile communication devices and portable apparatus has proliferated rapidly in recent years. A secondary battery has become the most significant power source for these devices, and hence, the battery charge system is also very important. Currently, the Li-ion battery has become the secondary battery with the most potential because it exhibits many advantages such as high energy density, no memory effect, high operation voltage, etc... [1]–[3]. Recent trends have shown that the Li-ion battery is widely used in portable apparatus and can also be used in aerospace, aircraft, and military applications [4]–[11]. To produce better battery charge system performance, many battery charge strategies have been proposed, such as the constant trickle current (CTC), constant current (CC), and CC and constant-voltage (CC–CV) charge strategies [12]. Among these strategies, the CTC charge strategy has a simple circuit structure without power elements with very low cost. However, the CTC charge strategy has a large charge time drawback. The charge time is longer than 10 h, making it an "overnight charger." To reduce the charge time, a current that is much larger than the trickle current is used in the charge system. This is called the CC charge strategy.

However, the drawback in the CC charge strategy is that a 100% precise fully charged detector is required. This device is not easy to implement. Therefore, undercharge and overcharge usually occur in the CC charge strategy. The CC–CV charge strategy was proposed to overcome this problem [12]. Under the CC–CV charge strategy, a CC is applied first until the battery voltage increases to a preset voltage (i.e., battery full charged voltage). This CV is held after reaching the preset voltage with the charge current automatically reduced. When the charge current reduces to zero, the battery is 100% fully charged. This charge strategy can effectively increase the battery charge speed, avoid overcharge, and achieve 100% fully charge. However, about 25%–40% of the complete charging period is needed to fulfill about 75%–80% of the totally required charge capacity during the CC mode. The time required for the remaining 20%–25% charge capacity in the CV mode should be three times the charge length in the CC mode [2]. Therefore, developing a novel charge strategy to replace the CV mode and reduce the charge time is a core problem in advanced battery charger systems.

Due to the complex electrochemical characteristics of a Li-ion battery, it is difficult to build a precise model for describing the Li-ion battery charge behavior [13]. The influential factors in the Li-ion battery charge voltage curve include the initial charged capacity, rate of charge current, rate of discharge current, temperature, depth of discharge, used cycle times, charge profile, overcharge, overdischarge, etc.... [1]–[3], [13]. While some charge voltage influential factors can be clearly described using mathematical models, most can only be described roughly. From the system point of view, the output response of a system, controlled by some certain factors and some uncertain factors, is a typical grey system [14], [15]. This system cannot be described and processed using conventional mathematical theories. In other words, general control theories cannot be used to design a perfect controller to improve or replace the CV mode. However, grey theory [14], [15], a system theory that can detail this kind of grey system, was recently proposed. The grey-predicted (GP) control technique was applied to various research areas [16]–[19]. In this paper, the grey theory is used to develop the GP Li-ion battery charge system (GP-LBCS) to replace the general CV mode using the proposed GP mode. The proposed GP-LBCS can instantly monitor and predict the charge behavior of the Li-ion battery for adaptively modifying its charge trajectories to decrease the charge time and maintain on the safe-charge conditions. A grey-controlled current source (GCCS) can adaptively provide a near-optimal charge current

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according to the present and future states of the Li-ion battery. In this paper, modeling, analysis, and design are conducted. A prototype of the GP-LBCS is also designed and implemented to assess the system performance. The experimental results show that the charge speed of the proposed GP-LBCS is 23% faster than the typical CC–CV charge system while maintaining safe-charge conditions.

II. Grey Model of Li-ION Battery

The study of grey theory was initiated in 1982 to overcome grey system problems [14], [15]. The gray theory ideal comes from the grey exponential law. By using the grey exponential law, we can have a near-optimum exponential curve to fit this sequence, and a predicted value can be obtained. Based on the aforementioned idea, the grey model is established from past information to the future based upon the past and present known or indeterminate information. This means that a grey model is built by only a few given numerical data without knowing the actual system or its mathematical model. The grey model is denoted by GM(m, n), where m is the order and n is the number of variables. The differential equation of the GM(m, n) is defined as

$$\frac{d^m x_1}{dt^m} + a_1 \frac{d^{m-1} x_1}{dt^{m-1}} + \cdots + a_{m-1} \frac{dx_1}{dt} + a_m x_1 = b_1 + b_2 x_2 + b_3 x_3 + \cdots + b_n x_n \quad (1)$$

where x_1 is the output variable, $x_2, x_3, \ldots, x_n$ are the input variables, and $a_1, a_2, \ldots, a_{m-1}$ and $b_1, b_2, \ldots, b_n$ are the coefficients that need to be determined. Obviously, GM(m, n) with large m can be used to predict variable x_1 with large dynamic behavior. Among the grey models, the GM(1, 1) model is the most simple and commonly used grey model. The GM(1, 1) model denotes a signal variable and the first-order linear dynamic model. However, the GM(1, 1) model can only deal with nonnegative sequences. Fortunately, the Li-ion battery voltage is always positive so that the GM(1, 1) model can be applied to the study of Li-ion battery charging forecast. The GM(1, 1) model for describing Li-ion battery can be directly obtained according to the following seven steps.

Step 1) Obtaining original battery voltage sequence $V^{(0)} = \{v^{(0)}(1), v^{(0)}(2), \ldots, v^{(0)}(n)\}$, where n is a number bigger than or equal to 4, and $v^{(0)}(1), \ldots, v^{(0)}(n)$ are battery voltages.

Step 2) Taking the accumulated generating operation (AGO) to get a nondecreasing voltage sequence described as

$$V^{(1)} = \left\{v^{(1)}(1), v^{(1)}(2), \ldots\ldots, v^{(1)}(n)\right\} \quad (2)$$

where

$$v^{(1)}(k) = \sum_{i=1}^{k} v^{(0)}(i), \qquad k = 1, 2, \ldots\ldots, n. \quad (3)$$

Step 3) Since $V^{(1)}$ is a nondecreasing voltage sequence, the exponential curve fitting method is used to predict $V^{(1)}$. According to (1), the fit exponential voltage function $\hat{v}^{(1)}(t)$ is described as

$$\frac{d\hat{v}^{(1)}(t)}{dt} + a\hat{v}^{(1)}(t) = b. \quad (4)$$

Therefore, the problem is to find the values of a and b so that

$$\hat{v}^{(1)}(t) = \left(v^{(0)}(1) - \frac{b}{a}\right) \cdot e^{-at} + \frac{b}{a} \quad (5)$$

is the best fitting voltage curve for $V^{(1)}$. Since $V^{(1)}$ is a discrete date sequence, $\hat{v}^{(1)}(t)$ and $(d\hat{v}^{(1)}(t))/dt$ in (4) are replaced by $(v^{(1)}(k) + v^{(1)}(k-1))/2$ and $v^{(1)}(k) - v^{(1)}(k-1) = v^{(0)}(k)$, respectively [15]. Therefore, (4) becomes the grey difference equation shown as

$$v^{(0)}(k) + aZ^{(1)}(k) = b \quad (6)$$

where

$$Z^{(1)}(k) = \left\{\frac{v^{(1)}(k) + v^{(1)}(k-1)}{2}\right\}. \quad (7)$$

Step 4) In order to determine the values of a and b, (6) is expressed as $B \cdot A = Y$ where

$$B = \begin{bmatrix} -Z^{(1)}(2) & 1 \\ -Z^{(1)}(3) & 1 \\ \vdots & \\ -Z^{(1)}(n) & 1 \end{bmatrix} \quad (8)$$

$$A = \begin{bmatrix} a \\ b \end{bmatrix} \quad (9)$$

$$Y = \left[\, v^{(0)}(2) \quad v^{(0)}(3) \quad \cdots \quad v^{(0)}(n) \,\right]^{\mathrm{T}}. \quad (10)$$

Step 5) By using the pseudoinverse method, the least root mean square error solution of A is

$$A = (B^{\mathrm{T}}B)^{-1}B^{\mathrm{T}}Y. \quad (11)$$

Step 6) Substituting the values of a and b obtained in (11) into (5), a predicted equation of $v^{(1)}$ is obtained and shown as

$$v^{(1)}(k+1) = \left[v^{(0)}(1) - \frac{b}{a}\right] e^{-ak} + \frac{b}{a}. \quad (12)$$

Step 7) Taking inverse-AGO of (12), the predicted Li-ion battery voltage $\tilde{v}$ can finally be obtained and described by

$$\tilde{v} = v^{(0)}(k+1) = \left(e^{-a} - 1\right) \left[v^{(0)}(1) - \frac{b}{a}\right] e^{-a(k-1)}. \quad (13)$$

In order to verify the Li-ion battery voltage predicting performance of the GM(1, 1) model, experiments for different charge current rates of 0.1, 0.3, and 0.5 C with 1-min sampling period

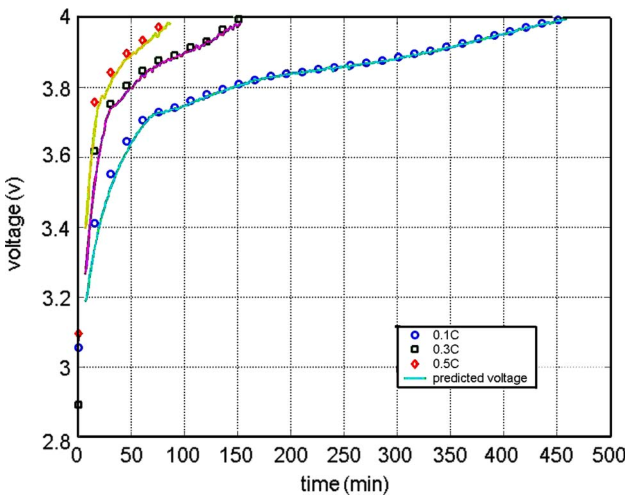


Fig. 1. GM(1, 1) predicted result in different charge current rates of 0.1, 0.3, and 0.5 C with 1-min sampling period.

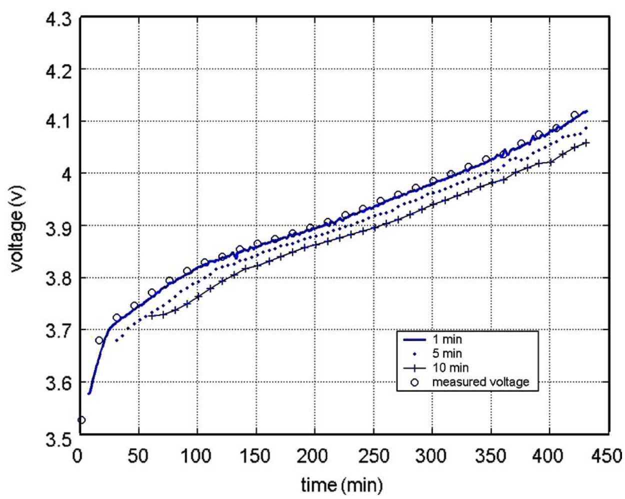


Fig. 2. GM(1, 1) predicted result in 0.1-C charge current with different sampling times of 1, 5, and 10 min.

are conducted and shown in Fig. 1. In the figure, the solid line "—" is the predicted battery voltage, and "○," "□," and "◊" are the measured battery voltages in 0.1-, 0.3-, and 0.5-C charge currents, respectively. It is clear from Fig. 1 that the GM(1, 1) model can actually predict the voltage curve of a Li-ion battery as we desired. To discuss the impact of sampling period on the predicting accuracy, experiments for 0.1-C charge current with different sampling periods of 1, 5, and 10 min are conducted and shown in Fig. 2. Fig. 2 shows that a larger sampling time results in a larger prediction error.

The aforementioned experiments show that GM(1, 1) model can precisely predict the Li-ion battery voltage curve provided under a sufficiently small sampling period. To precisely predict the Li-ion battery voltage curve, the safe sampling period T_s, which is defined as the maximum sampling period, needed to be determined. In GP-LBCS, the sampling period should be smaller than the safe sampling period T_s to make sure that the GP-LBCS works in safe-charge area (SCA) all the time. Otherwise, the overcharge state will appear to reduce the life cycle of a Li-ion battery.

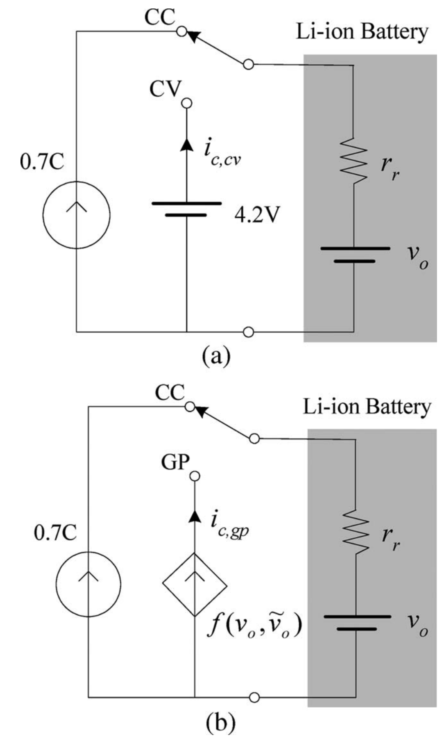


Fig. 3. Equivalent charge models for one-cell Li-ion battery: (a) CC–CV charge strategy and (b) proposed GP charge strategy.

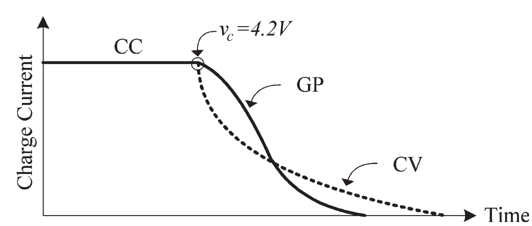


Fig. 4. Predicted charge trajectories of the CC–CV (—) and proposed GP (—) charge processes for one Li-ion battery.

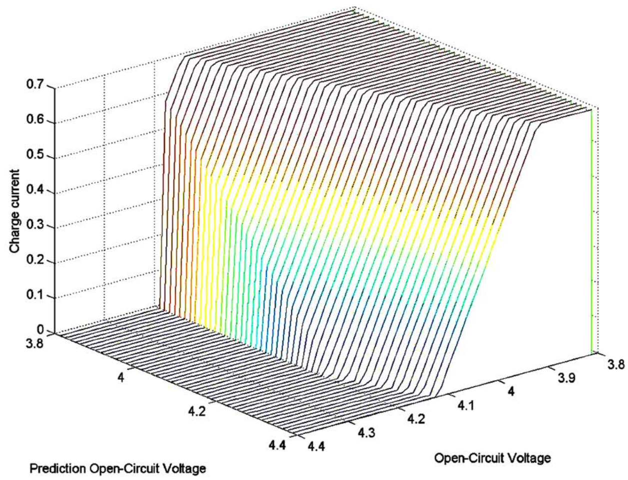
III. GP Charge Strategy

The proposed GP and the typical CC–CV charge systems for one Li-ion battery are shown in Fig. 3(a) and (b), respectively, in which r_r is the inter-pack-resistance, including the resistances of the protection FET, the thermal fuse, the PCB routes, and connectors. In Fig. 3(a) and (b), the CC mode in each charger is the same. The GP mode is proposed to replace the general CV mode so as to proceed with the charge process in safe conditions and reduce the charge time. The predicted charge trajectories of the two charge systems are shown in Fig. 4 for comparison. Remarkably, the charge performances are the same in both CC modes, but the charge speed in the GP mode is obviously faster than that in the general CV mode.

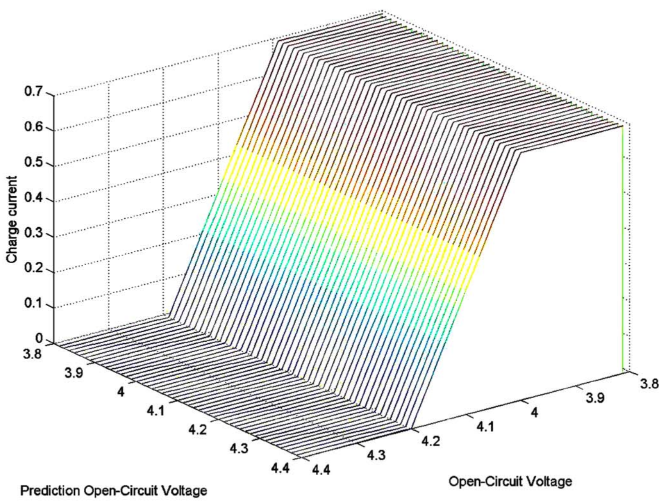
In the proposed GP charge strategy, the GP charge current $i_{\mathrm{c,gp}}$ in GP mode is defined as

$$i_{\mathrm{c,gp}} = \mathrm{MIN}\left(0.7\ \mathrm{C}, \frac{4.2 - v_o}{r_r} + \beta \cdot \frac{(4.2 - \tilde{v}_o)}{r_r}\right) \qquad (14)$$

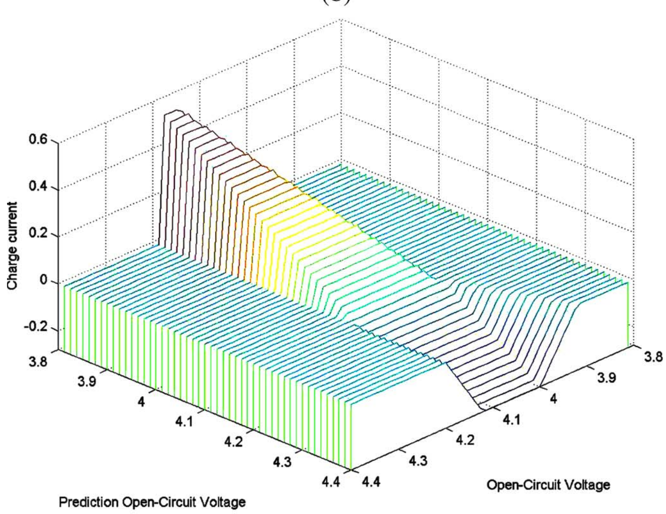
where β is the enhance factor. It is noticeable that the GP charge current $i_{\mathrm{c,gp}}$ is always smaller than or equal to 0.7 C. In a special case of $\beta = 0$, the GP charge current $i_{\mathrm{c,gp}}$ of the



(a)



(b)



(c)

Fig. 5. Simulated charge trajectories of (a) the proposed GP-LBCS, (b) the CC–CV charge system, and (c) the difference between GP-LBCS and CC–CV.

proposed GP mode becomes $i_{\mathrm{c,gp}} = (4.2 - v_o)/r_r$, which is the same as the CV charge current $i_{\mathrm{c,cv}} = (4.2 - v_o)/r_r$ in CV mode of a CC–CV charge strategy. Fig. 5(a) shows the 3-D Y-mesh diagrams for describing the charge trajectories of the proposed GP charge strategy with $\beta = 0.4$ in simulation. For analysis convenience, the charge trajectories of the general CC–CV charge strategy are also simulated in Fig. 5(b). Comparing Fig. 5(a) and (b), one can find that the GP charge current $i_{\mathrm{c,gp}}$ of the proposed GP charge strategy is increased to decrease the battery charge time when the predicted open-circuit voltage $\tilde{v}_o$ is smaller than 4.2 V (i.e., battery full charged voltage). On the other hand, the GP charge current $i_{\mathrm{c,gp}}$ is decreased to avoid battery overcharge when the predicted open-circuit voltage $\tilde{v}_o$ is larger than 4.2 V. The increase/decrease charge current in GP-LBCS is shown in Fig. 5(c) to show the effect of the predict-control battery charge method.

IV. GP-LBCS

Fig. 6(a) shows a block diagram of the proposed GP-LBCS, which consists of a grey prediction controller (GPC), a GCCS, a timer, a switch, and a voltage sensor and converter (VSC). The proposed GP-LBCS is operated in SCA, which is defined as a safe maximum in-charging voltage of the Li-ion battery [20]. The proposed GP-LBCS working process is divided into four working states: sense state (SS), CC charge state (CCCS), calculation state (CS), and prediction charge state (PCS) with the sense period T_{SS}, charge period T_{CC}, and calculation period T_{CS} as defined in Fig. 6(b), in which the CS and the PCS construct the GP mode and the CCCS conducts the CC mode. After power on, the GP-LBCS directly goes into SS. In SS, the primary work is to measure the open-circuit voltage v_o and in-charging voltage v_c for checking the state of charge in the Li-ion battery and then determine the next GP-LBCS working state. If the sensed open-circuit voltage v_o is bigger than or equal to 4.2 V, the GP-LBCS will go into end state and finish a battery charge process. Otherwise, the GP-LBCS will go into the PCS or CCCS according to sensed in-charging voltage v_c. If the sensed in-charging voltage v_c is smaller than 4.2 V, the GP-LBCS works in the CCCS (i.e., CC mode) to charge the Li-ion battery with a constant charge current for a constant charge period T_{CC}. Otherwise, the GP-LBCS goes into the CS to calculate (13) and (14) to obtain the GP charge current $i_{\mathrm{c,gp}}$.

After CS, the GP-LBCS will go into the PCS to charge the Li-ion battery using the calculated GP charge current $i_{\mathrm{c,gp}}$ for a constant charge period T_{CC}. Finally, GP-LBCS will go back to SS to sense the state of the battery again. Remarkably, the charge period T_{CC} is much bigger than the sense period T_{SS} and the calculation period T_{CS}, and the minimum calculation period T_{CS} is determined by computing the speed of the microprocessor. For more clearly explaining the working process of the proposed GP-LBCS, the processes in SS, CS, PCS, and CCCS are separately described as follows.

A. SS

In SS, the VSC first measures the in-charging voltage v_c of the Li-ion battery and transfers the in-charging voltage v_c into a digital code N_{Vc} during switch on. Next, the GPC receives the digital code N_{Vc} and turns the switch off. At this time, the Li-ion battery is floated from the charge circuit. The open-circuit voltage v_o of the Li-ion battery is sensed and transferred

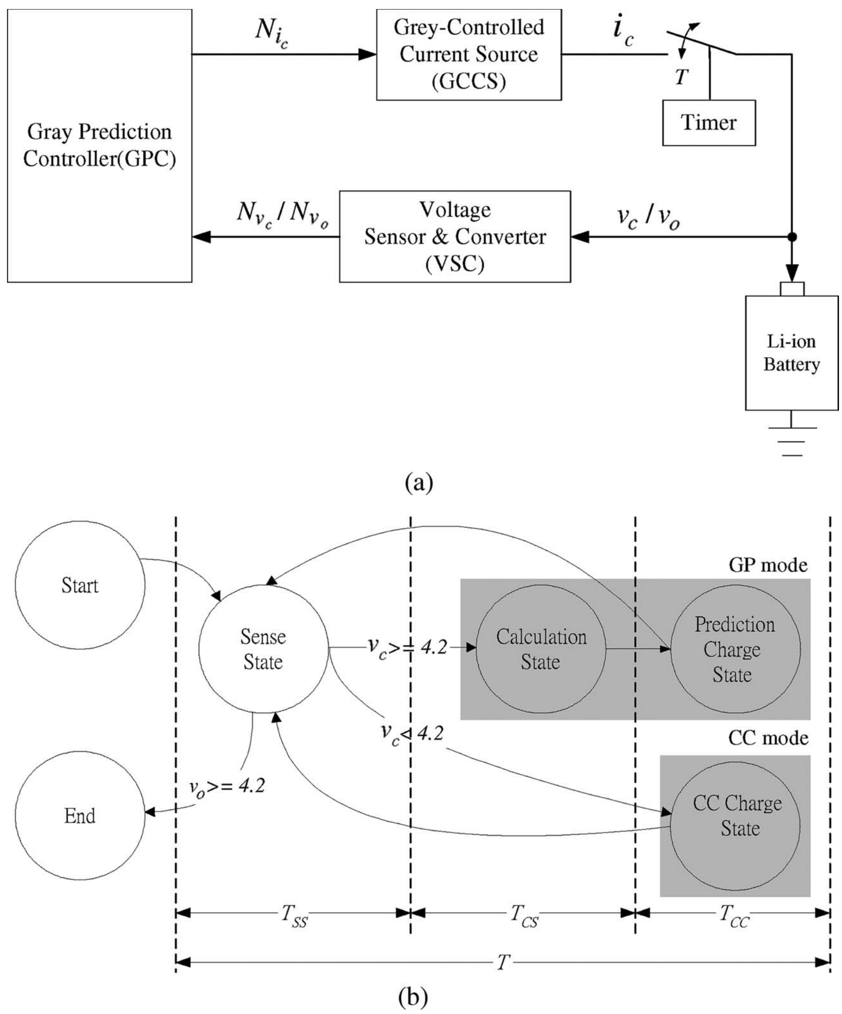


Fig. 6. (a) Block diagram and (b) the state diagram of the proposed GP-LBCS.

into a digital code N_{Vo} by VSC. The digital code N_{Vo} is then sent to the GPC. After that, the GPC can determine the next working state according to the received digital codes N_{Vc} and N_{Vo}.

B. CS

In CS, (13) is processed to predict the Li-ion battery voltage of the next sampling time. The GP charge current digital code $N_{i_{\mathrm{c,gp}}}$ is then obtained according to (14).

C. PCS and CCCS

In PCS, GPC sends the digital code $N_{i_{\mathrm{c,gp}}}$ obtained in CS to the GCCS to generate the desired GP charge current $i_{\mathrm{c,gp}}$ to charge the Li-ion battery. At the same time, the timer counts to control the Li-ion battery charging for a constant charge period T_{CC}. Similarly, the GPC commands the GCCS to generate a CC (i.e., 0.7 C) and then to charge the Li-ion battery for a charge period T_{CC} in CCCS.

V. Design Considerations

The sampling period T and the enhance factor β are the key factors in designing the proposed GP-LBCS. The sampling period T, shown in Fig. 6(b), consists of the sense period T_{SS}, calculation period T_{CS}, and charge period T_{CC}. The charge period T_{CC} is for charging Li-ion battery in PCS. Two samples conducted in SS during period T_{SS}, including the detection of v_o in T_{SS1} and v_c in T_{SS2}. The sense period $T_{\mathrm{SS}} = T_{\mathrm{SS1}} + T_{\mathrm{SS2}}$ is much smaller than the charge period T_{CC}. The calculation period T_{CS} is primarily determined by the microprocessor operating speed and is also much smaller than the charge period T_{CC}. For convenience in design, we presume that $T \cong T_{\mathrm{CC}}$. The relation between the sampling period T and battery charge current i_c for working in the SCA is presented and shown in the following inequality [20], [21]:

$$\frac{i_c}{C_b} \cdot T < 0.05 \tag{15}$$

where C_b is the equivalent capacitance. If a measurement error is considered in (15), (15) can be rewritten as

$$\frac{i_c \cdot T}{C_b} < 0.05 - |E| \tag{16}$$

where $|E|$ is the absolute value of the measurement error. In this paper, the prediction error E_p is treated as the measurement error; therefore, (16) can be directly rewritten as

$$\frac{i_c \cdot T}{C_b} < 0.05 - |E_p| . \tag{17}$$

From (17), an inequality, as shown in (18), can be obtained to determine the sampling period region, which guarantees that GP-LBCS will work in SCA.

$$T < \frac{0.05 \cdot C_b - C_b \cdot |E_p|}{i_c}. \tag{18}$$

The safe sampling period T_s can thus be obtained and shown as

$$T_s = \frac{0.05 \cdot C_b - C_b \cdot |E_p|}{i_c}. \tag{19}$$

In practical applications, the maximum prediction error E_{mp} and the maximum charge current $i_{c,\max}$ should be used as the prediction error E_p and charge current i_c in (19) for safe consideration. Therefore, the formula used to determine the safe sampling period T_s is obtained and shown as follows:

$$T_s = \frac{0.05 \cdot C_b - C_b \cdot |E_{\mathrm{mp}}|}{i_{c,\max}}. \tag{20}$$

According to (20), we can decide a suitable sampling period T to guarantee that the designed GP-LBCS is working in SCA all the time. For safe-charge consideration, the in-charging voltage v_c on the Li-ion battery should not be over the maximum fault over voltage threshold (i.e., 4.25 V) for one Li-ion cell [22], [23], which is written as

$$v_c = i_{c,\mathrm{gp}} \cdot r_r + v_o \le 4.25. \tag{21}$$

From (14) and (21), we have

$$\beta \cdot (4.2 - \tilde{v}_o) \le 0.05. \tag{22}$$

Since $\tilde{v}_o = \tilde{v}_c - i_{c,\mathrm{gp}} \cdot r_r$ and $\tilde{v}_c \cong v_c$, enhance factor β can then be estimated by

$$\beta \le \frac{0.05}{i_{c,\mathrm{gp}} \cdot r_r}. \tag{23}$$

Therefore, the suitable enhance factor β value should satisfy the following inequality to guarantee that the designed GP-LBCS will work in SCA:

$$\beta < \frac{0.05}{i_{c,\max} \cdot r_r}. \tag{24}$$

VI. Design Example

In this section, we examine a design example of the proposed GP-LBCS for a 600-mAh Li-ion battery. For safe-charge consideration, the maximum permissible charge current is specified as 0.42 A (i.e., $i_{c,\max} = 0.42$ A $= 0.7$ C), where 1 C $= 0.6$ A is the rate charge current for the 600-mAh Li-ion battery. Fig. 7 shows the physical block diagram of the proposed GP-LBCS. The realized GP-LBCS consists of an 8-b microprocessor EM78459 with a built-in analog-to-digital converter, a digital-to-analog converter DAC0800, a variable current generator (VCG), a switch, and a Li-ion battery. By experiments, the maximum prediction error $E_{\mathrm{mp}} = 0.03$, the equivalent capacity of this Li-ion battery is estimated as $C_b = \Delta Q/\Delta V_o \cong 1430$ F, and its $r_r \cong 0.2\ \Omega$. Since the maximum charge current $i_{c,\max} = 0.42$ A, the maximum prediction error $E_{\mathrm{mp}} = 0.03$, and the inter-package-resistance $r_r \cong 0.2\ \Omega$, from (20) and (24), the safe sampling period T_s and enhance factor β can be obtained as 68 s and 0.6, respectively. In this design, sampling period T and the enhance factor β are, respectively, chosen as 60 s and 0.4 for safe consideration.

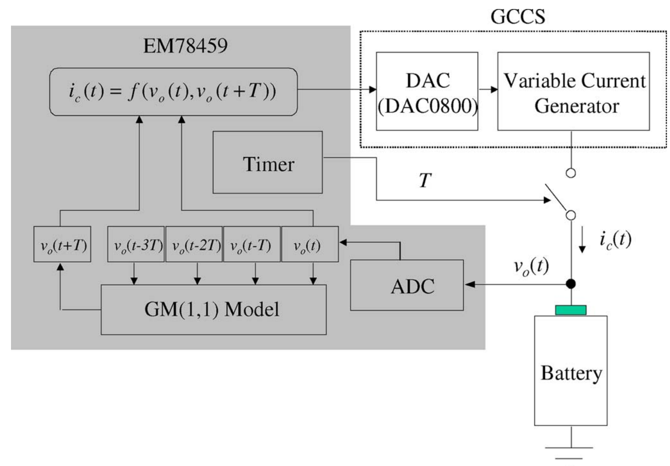


Fig. 7. Block diagram of the realized GP-LBCS.

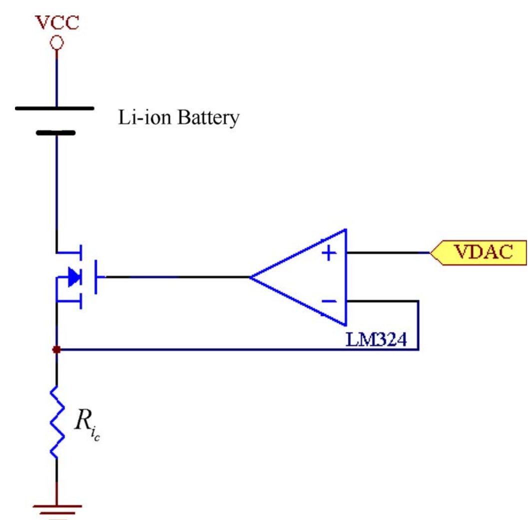


Fig. 8. VCG circuit.

The VCG circuit used in this design example is shown in Fig. 8. The VCG is constructed by an operation amplifier LM324, a power MOSFET, and a resistor R_{i_c}. The output charge current of the VCG is directly obtained using the following equation:

$$i_c = \frac{V_{\mathrm{DAC}}}{R_{i_c}} \tag{25}$$

where V_{DAC} is the output voltage of the DAC0800 and controlled by the microprocessor EM78459. Therefore, we can get a desired charge current trajectory by programming the software in the microprocessor EM78459.

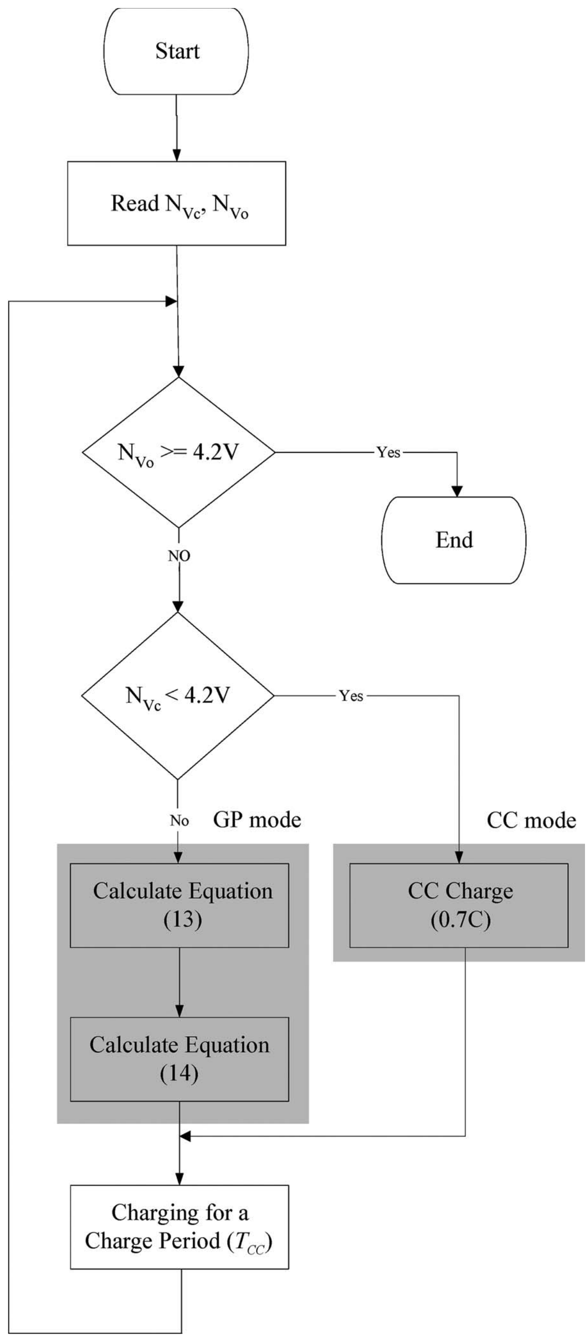


Fig. 9. GPC software flowchart.

The flowchart of the software in the microprocessor EM78459 is shown in Fig. 9. First, the in-charging voltage digital code N_{Vc} and the open-circuit voltage digital code N_{Vo} of the Li-ion battery are read. The microprocessor EM78459 then detects whether the Li-ion battery is fully charged according to the open-circuit voltage digital code N_{Vo}. If the open-circuit voltage digital code N_{Vo} is larger than or equal to 4.2 V, the Li-ion battery is fully charged and the battery charge process will be ended. Otherwise, the Li-ion battery will be charged by CC mode or GP mode according to the in-charging voltage digital code N_{Vc}. If the in-charging voltage digital code N_{Vc} is smaller than 4.2 V, the microprocessor EM78459 will send a 0.7-C CC command to GCCS to generate a 0.7-C CC to charge the Li-ion battery for a charge period T_{CC}. If the in-charging voltage digital code N_{Vc} is bigger than or equal to 4.2 V, (13) and (14) are sequentially calculated to get the GP charge current digital code $N_{i_{\mathrm{c,gp}}}$. Finally, the GCCS produces the desired GP charge current $i_{\mathrm{c,gp}}$ to charge the Li-ion battery. After several cycles as mentioned above, the Li-ion battery will be fully charged.

VII. Experimental Results

Experimental results of the proposed GP-LBCS with $\beta = 0.4$ and conventional CC–CV charge system for 600-mAh Li-ion battery are shown in Fig. 10. In order to measure the battery charging data (e.g., battery voltage, charge current, and integration of charge current), the battery charge measurement topology proposed by Wang and Stuart [24] is referred. Fig. 10(a) shows the in-charging voltage v_c with respect to the charge time. Fig. 10(a) shows that the in-charging voltage curves in both the proposed GP-LBCS and the conventional CC–CV charge system are always smaller than 4.25 V. That means both the proposed GP-LBCS and the conventional CC–CV charge system do not have an overcharge situation that impacts the battery life cycles [25]. Fig. 10(b) shows the open-circuit voltage v_o with respect to the charge time. According to Fig. 10(b), we can see that the open-circuit voltage curves almost reach to 4.2 V at the end. This means that both the proposed GP-LBCS and the general CC–CV charge current can fully charge the Li-ion battery. From Fig. 10(a) and (b), we can also see that the open-circuit voltage v_o is always less than the in-charging voltage v_c until the battery is fully charged in both the proposed GP-LBCS and the conventional CC–CV charge system. Fig. 10(c) shows the charge current with respect to the charge time. Obviously, the charge current in the GP mode is greater than that in the CV mode before the Li-ion battery is fully charged. Thus, the charge times for the proposed GP-LBCS and the general CC–CV charge system are 132 and 188 min, respectively. The charge speed of the proposed GP-LBCS has been improved to about 29.8% compared with that of the general CC–CV charge system. Remarkably, the charge times in the CC mode for both of the proposed GP-LBCS and the conventional CC–CV charge system are the same, i.e., 81 min. The charge times in the proposed GP mode and the general CV mode are 51 and 107 min, respectively. This indicates that the charge time in the proposed GP mode has been reduced to about 52% compared with that in the general CV mode. Fig. 10(d) shows the charge energies with respect to the charge time. From Fig. 10(d), it is evident that the total charge energies for the proposed GP-LBCS and the general CC–CV charge system are 2856.2 and 2902.5 mWh, respectively. The charge energy of the proposed GP-LBCS has been reduced to about 1.6% compared with that of the general CC–CV charge system. This indicates that battery charge efficiency is improved to about 1.6% by the proposed GP battery charge strategy. Note that the charge energies in the CC mode for both the proposed GP-LBCS and the general CC–CV are the same, i.e., 2247.3 mWh. Thus, the charge energies for the proposed GP mode and the general CV mode are 608.9 and 655.2 mWh, respectively. This means that the charge efficiency of the proposed GP mode is improved to about 7.1% compared with that of the general CV mode.

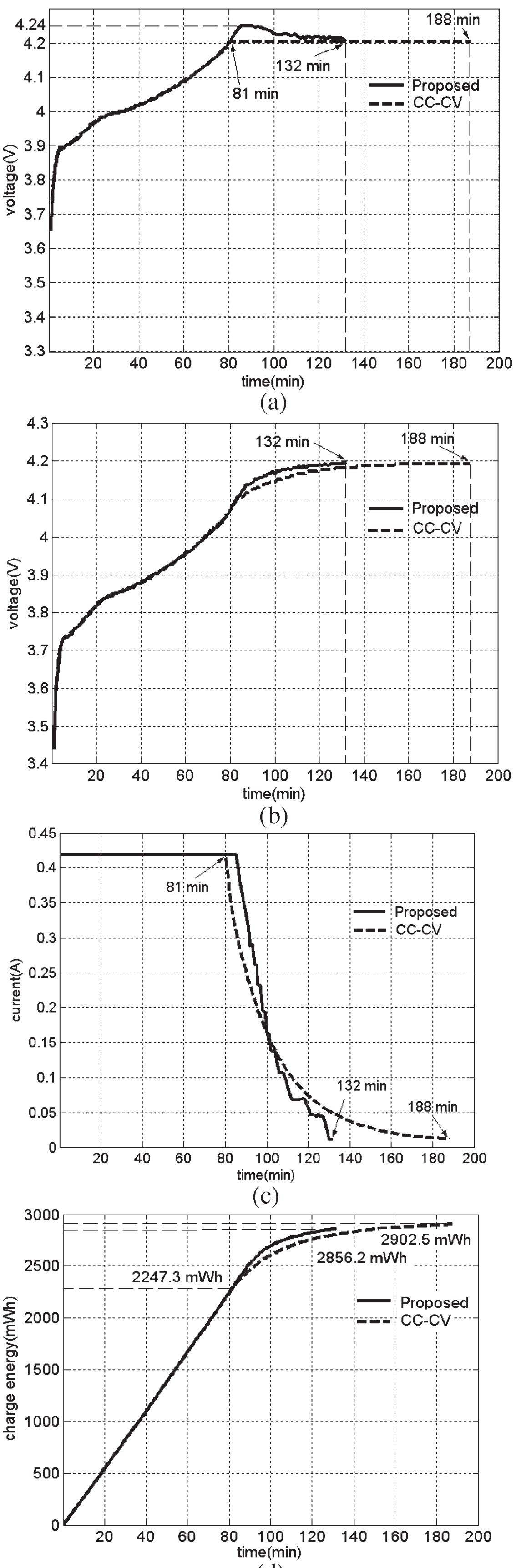


Fig. 10. (a) Charge current, (b) the in-charging voltage, (c) the open-circuit voltage, and (d) the charge energy of the CC–CV charge system (—) and the proposed GP-LBCS (—) with $\beta = 0.4$ for 600-mAh Li-ion battery.

The developed GP-LBCS for 600-mAh Li-ion battery in Section VI is modified to charge a 1000-mAh Li-ion battery by resetting $\beta = 0.1$, $T = 60$ s, and $i_{c,\max} = 0.7$ A (i.e., 0.7 C). Experimental results from the proposed GP-LBCS with $\beta = 0.1$ and conventional CC–CV charge system for 1000-mAh Li-ion battery are shown in Fig. 11. From Fig. 11(a) and (b), we can see that the Li-ion battery is fully charged without overshoot in the proposed GP-LBCS for charging a 1000-mAh Li-ion battery. From Fig. 11(c), we also see that the charge times of the proposed GP-LBCS and the general CC–CV charge system are 181 and 238 min, respectively. This indicates that the charge speed of the proposed GP-LBCS has been improved to about 23.9% compared with that of the general CC–CV charge system. Remarkably, the charge times in the CC mode for both of the proposed GP-LBCS and the conventional CC–CV are the same, i.e., 70 min. The charge times in the proposed GP mode and the general CV mode are 111 and 168 min, respectively. This indicates that the charge time in the proposed GP mode has been reduced to about 34% compared with that in the general CV mode. From Fig. 11(d), we see that the charge energies of the proposed GP-LBCS and the general CC–CV charge system are 4999.8 and 5291.2 mWh, respectively. The charge energy of the proposed GP-LBCS has been reduced to about 5.5% compared with that of the general CC–CV charge system. This indicates that battery charge efficiency is improved to about 5.5% by using the proposed GP battery charge strategy. The charge energies in the CC mode for both the proposed GP-LBCS and the general CC–CV are the same, i.e., 3278.2 mWh. The charge energies for the proposed GP mode and the general CV mode are 1721.6 and 2013 mWh, respectively. This means that the charge energy of the proposed GP mode is reduced to about 14.4% compared with that of the general CV mode.

According to the aforementioned two experiments, we can conclude that the proposed GP-LBCS can increase the charge speed above 23% and improve the charge efficiency above 1.6% compared with the general CC–CV charge system. In addition, the developed GP model can increase the charge speed above 34% and improve the charge efficiency above 7% compared with the general CV mode. The developed GP-LBCS prototype is designed for a single Li-ion battery; however, it also can charge series-connected Li-ion batteries using a voltage divider. An equalizer [26]–[28] is strongly recommended for use in a series-connected Li-ion battery charge system to increase the safety. Since other kinds of secondary batteries, such as Ni–Ca, Ni–HM, and Lead acid batteries, are also grey systems, the developed GP charging technique is suitable for Ni–Ca, Ni–HM, and Lead acid batteries.

VIII. Conclusion

A Li-ion battery was viewed as a grey system, and a GM(1, 1) model was employed to predict the Li-ion battery charge behavior. By using the GP technique, a GP-LBCS was successfully implemented. The proposed GP-LBCS was designed to replace the general CV mode using the GP mode for improving the Li-ion battery charge behavior. A Li-ion battery grey model was conducted for periodically predicting the charge behavior. A GP algorithm was built in GP-LBCS to speed up the charge

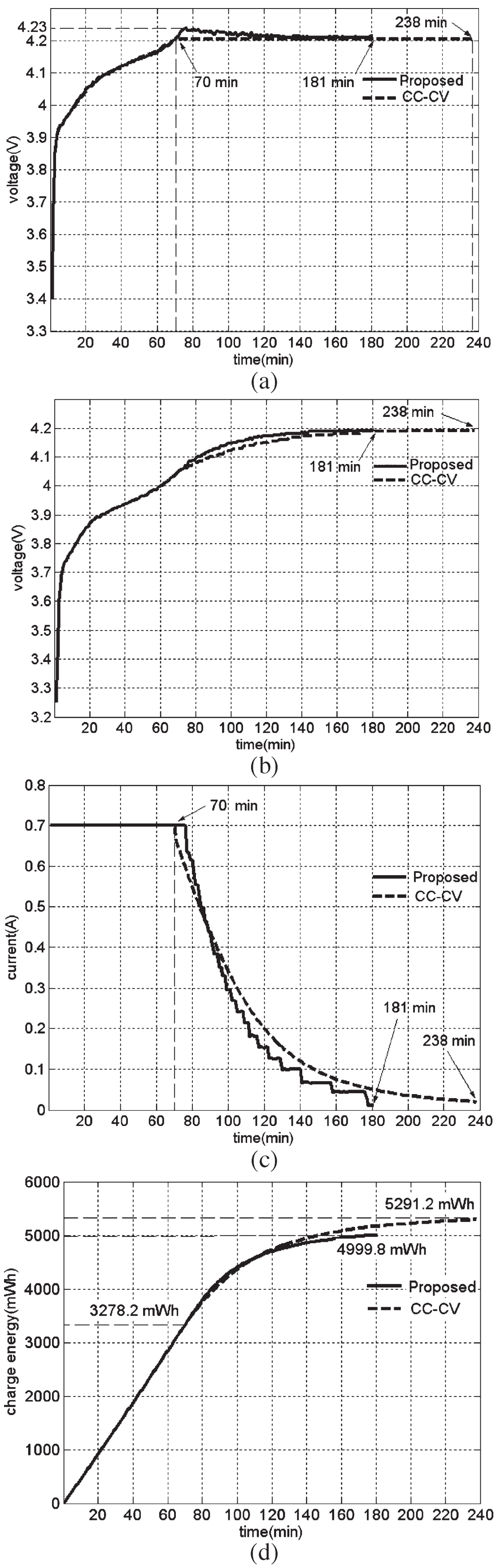


Fig. 11. (a) Charge current, (b) the in-charging voltage, (c) the open-circuit voltage, and (d) the charge energy of the CC–CV charge system (—) and the proposed GP-LBCS (—) with $\beta = 0.1$ for 1000-mAh Li-ion battery.

trajectories faster and maintain the charge trajectory in the SCA. A 3-D Y-mesh diagram was explored to describe the charge trajectories of the charge behavior. A GP-LBCS prototype was designed and implemented to assess the charge performance. The proposed GP-LBCS, compared with the general CC–CV charge system, can improve the charge speed above 23% and the battery charge efficiency above 1.6%. The charge speed and the charge efficiency in the proposed GP mode were improved above 34% and 7%, respectively, compared with that in the general CV mode.

References

[1] G. Nagasubramanian and R. G. Jungest, "Energy and power characteristics of lithium-ion cell," *J. Power Sources*, vol. 72, no. 2, pp. 189–193, Apr. 1998.

[2] J. A. Carcone, "Performance of lithium-ion battery systems," in *Proc. WESCON Idea/Microelectron. Conf. Rec.*, Anaheim, CA, Sep. 1994, pp. 242–248.

[3] B. Carter, J. Matsumoto, A. Prater, and D. Smith, "Lithium ion battery performance and charge control," in *Proc. 31st Intersoc. Energy Convers. Eng. Conf.*, Washington, DC, Aug. 1996, pp. 363–368.

[4] L. R. Chen, J. J. Chen, N. Y. Chu, and G. Y. Han, "Current-pumped battery charger," *IEEE Trans. Ind. Electron.*, vol. 55, no. 6, pp. 2482–2488, Jun. 2008.

[5] L. R. Chen, "A design of an optimal pulse charge system by frequency-varied technique," *IEEE Trans. Ind. Electron.*, vol. 54, no. 1, pp. 398–405, Feb. 2007.

[6] L. R. Chen, C. C. Huang, and S. L. Wu, "A design of a Li-ion battery duty-varied pulse charger," in *Proc. 33rd Annu. Conf. IEEE Ind. Electron. Soc.*, Nov. 2007, pp. 402–407.

[7] S. J. Lee, J. H. Kim, J. M. Lee, and B. H. Cho, "The state and parameter estimation of an Li-ion battery using a new OCV-SOC concept," in *Proc. IEEE Power Electron. Spec. Conf.*, Jun. 2007, pp. 2799–2803.

[8] H. C. Yang and L. R. Dung, "An accurate Lithium-ion battery gas gauge using two-phase STC modeling," in *Proc. IEEE Int. Symp. Ind. Electron.*, Jun. 2007, pp. 866–871.

[9] M. C. Smart, B. V. Ratnakumar, L. D. whitcanack, K. B. Chin, S. Surampud, and S. Byers, "Lithium-ion batteries for aerospace," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 19, no. 1, pp. 18–25, Jan. 2004.

[10] G. C. Bruce and L. Marcoux, "Large lithium ion batteries for aerospace and aircraft applications," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 16, no. 9, pp. 24–28, Sep. 2001.

[11] H. Oman, "Aerospace and military battery applications," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 17, no. 10, pp. 29–35, Oct. 2002.

[12] R. C. Cope and Y. Podrazhansky, "The art of battery charging," in *Proc. 14th Battery Conf. Appl. Adv.*, Long Beach, CA, Jan. 1999, pp. 233–235.

[13] S. Gold, "A PSPICE macromodel for Lithium-ion batteries," in *Proc. IEEE 12th Battery Conf. App. Adv.*, Long Beach, CA, Jan. 1997, pp. 215–222.

[14] J. Deng, "Control problem of grey system," *Syst. Control Lett.*, vol. 5, pp. 288–294, 1982.

[15] J. Deng, "Introduction to grey system theory," *J. Grey Syst.*, vol. 1, no. 1, pp. 1–24, 1989.

[16] S. L. Su, Y. C. Su, and J. F. Huang, "Grey-based power control for DS-CDMA cellular mobile systems," *IEEE Trans. Veh. Technol.*, vol. 49, no. 6, pp. 2081–2086, Nov. 2000.

[17] R. C. Luo and T. M. Chen, "Autonomous mobile target tracking system based on grey-fuzzy control algorithm," *IEEE Trans. Ind. Electron.*, vol. 47, no. 4, pp. 920–931, Aug. 2000.

[18] S. J. Huang and C. L. Huang, "Control of an inverted pendulum using grey prediction model," *IEEE Trans. Ind. Appl.*, vol. 36, no. 2, pp. 452–458, Mar./Apr. 2000.

[19] R. J. Wai and C. H. Tu, "Adaptive grey control for hybrid resonant driving linear piezoelectric ceramic motor," *IEEE Trans. Ind. Electron.*, vol. 53, no. 2, pp. 640–656, Apr. 2006.

[20] G. C. Hsieh, L. R. Chen, and K. S. Huang, "Fuzzy-controlled Li-ion battery charge system with active state-of-charge controller," *IEEE Trans. Ind. Electron.*, vol. 48, no. 3, pp. 585–593, Jun. 2001.

[21] L. R. Chen, C. H. Lin, R. C. Hsu, B. G. Ku, and C. S. Liu, "A study of Li-ion battery charge forecasting using grey theory," in *Proc. IEEE INTELEC*, Oct. 2003, pp. 745–749.

[22] *Product Data Handbook*, Unitrode Corp., Merrimack, NH, 1997.

[23] L. W. Hruska, "Smart battery and Lithium-ion voltage profiles," in *Proc. IEEE 12th Annu. Battery Conf. Appl. Adv.*, Jan. 1997, pp. 205–210.
[24] X. Wang and T. Stuart, "Charge measurement circuit for electric vehicle batteries," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 38, no. 4, pp. 1201–1209, Oct. 2002.
[25] S. P. Roland, "A dynamic voltage-compensation technique for reducing charge time in lithium-ion batteries," in *Proc. 15th Annu. Battery Conf. Appl. Adv.*, Jan. 2000, pp. 179–184.
[26] Y. S. Lee and M. W. Cheng, "Intelligent control battery equalization for series connected lithium-ion battery strings," *IEEE Trans. Ind. Electron.*, vol. 52, no. 5, pp. 1297–1307, Oct. 2005.
[27] Y. S. Lee and G. T. Chen, "Quasi-resonant zero-current-switching bidirectional converter for battery equalization applications," *IEEE Trans. Power Electron.*, vol. 21, no. 5, pp. 1213–1224, Sep. 2006.
[28] A. C. Baughman and M. Ferdowsi, "Double-tiered switched-capacitor battery charge equalization technique," *IEEE Trans. Ind. Electron.*, vol. 55, no. 6, pp. 2277–2285, Jun. 2008.

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